Jiahang Bai

RESEARCH CURRICULUM VITAE

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I'm an undergraduate at Jilin University's School of Artificial Intelligence and spent nearly a year working with robots as an algorithm intern at Dexmal
One question keeps me curious: beyond scaling data and parameters, how can we expand the context robots can use to act, from current observations to past interactions?
As foundation models like GPT Astra become more capable, I see a key step toward physical AI in pairing strong vision-language models with action token representations that preserve what matters for control
My first research project, S²Mem, explores how robots can retain fleeting visual cues and use them to guide later actions

Education

Jilin University 2024–Present

Undergraduate in Artificial Intelligence | School of Artificial Intelligence

GPA 3.63 / 4.00

Selected coursework Probability Theory and Mathematical Statistics A (94/100), Deep Learning (91.1/100), Machine Learning and Pattern Recognition II (92/100), Frontiers of Artificial Intelligence I (96/100), Data Structures and Algorithm Analysis II (91/100), and Evolutionary Computation (90.6/100)

Research and Industry Experience

Dexmal Jan–Nov 2026

Embodied AI Algorithm Intern | Mentor: Ze Chen

- Debugged robot setups and ran multiple manipulation experiments on Mobile ALOHA and Dexmal DOS-W1
- **Jan–Feb · Action generation** Explored step distillation for π_0 to reduce the number of iterative denoising steps required to generate actions
- **Mar–Apr · High-level reasoning** Explored how the pretrained DM05 vision-language backbone could support embodied chain-of-thought (ECoT) by using visual observations to infer task progress and choose the next subgoal before generating low-level actions
- **May–Nov · Memory for VLA** Developed S²Mem as first author, using proprioceptive motion to select visual events for long-term memory and training memory queries to recover DINOv2 features of masked past events, with evaluation in simulation and on real robots
- **Aug–Nov · Collaborative research** Contributed to tactile VLA for contact-rich manipulation (FutureTacVLA) and online reinforcement learning for VLA post-training (Sight-RLT)

HKUST (Guangzhou) Jul–Aug 2026

Research Intern (Remote) | Mentors: Qiwei Liang and Renjing Xu

- Worked on representation learning and multi-task policy learning for world action models

CAS Undergraduate Innovation and Research Training Program Dec 2024–Jun 2025

Research Intern (Remote) | Mentor: Jianghao Wang

- Worked on BERT-based sentiment analysis and spatiotemporal social media data processing at the Institute of Geographic Sciences and Natural Resources Research, CAS

Jilin University Dec 2025–Present

Undergraduate Researcher | Mentor: Yixing Gao

- Reproduced OpenVLA and $\pi_{0.5}$, fine-tuned policies for real-robot shirt folding, and explored whether policies could retry after failed attempts
- Worked with Baxter, RealMan robots, and LimX Dynamics TRON 2

Research Interests

Memory and sequence modeling for vision-language-action (VLA) policies, with a focus on learning action-relevant representations for long-horizon robotic manipulation

Manuscripts Under Review

S²Mem: From Sparse Cues to Decision-Relevant Memory in Vision-Language-Action Models

Jiahang Bai, Leyang Wang, Zixuan Huang, Wenxuan Guo, Xutao Sun, Kaixi Ying, Xiaowen Qin, Ze Chen, Erjin Zhou, Yixing Gao | *First author*, *2026*

Uses robot motion to retain brief visual cues in multi-scale memory and trains the policy to recover masked past events
Achieved 95.3% on RoboTwin-MeM and 79.6% on RMBench (+20.1 and +11.8 percentage points over EventVLA), and 83.33% mean success on real-robot tasks (+35.0 points over the strongest baseline)

Sight-RLT: Control-Oriented RL Tokens from Frozen Vision-Language-Action Models

Zixuan Huang, Yanxiang Ma, Jianyuan Guo, Jiaqi Liu, Leyang Wang, **Jiahang Bai**, Ze Chen, Kaixin Liu, Erjin Zhou, Chang Xu | *Co-author*, *2026*

Combines action-supervised queries with pooled features from a frozen VLA model through late fusion in the actor and critics for online reinforcement learning
On ManiSkill-Fine, improved mean success over RLT by 11.27 percentage points on seen initializations and 8.54 points on unseen initializations

FutureTacVLA: Dynamics-Aware In-Expert Foresight for Contact-Rich Robotic Manipulation

Leyang Wang, Yi Wang, **Jiahang Bai**, Zixuan Huang, Peili Ma, Ruitao Zhang, Ze Chen, Erjin Zhou, Jikai Wang | *Co-author*, *2026*

Integrates visual and tactile predictions into the action expert and learns contact dynamics from force and force variation
Achieved 82.50% mean success across six UniVTAC tasks (+15.84 percentage points over FTP-1) and outperformed baselines on four real-robot tasks

Selected Competitions

RoboCup China Humanoid League 2026

Team Captain | National Second Prize and Top 8 Nationwide

- Designed and developed core algorithms for robot positioning and return-to-field behavior, then used feedback from real-robot trials to tune movement speed and improve shooting performance
- Coordinated the team, organized practice and task assignments, and served as the point of contact with competition organizers

Mathematical Contest in Modeling 2025

Team Captain | Finalist

- Led problem decomposition and model formulation for Problem C and designed and implemented core algorithms for Olympic gold-medal forecasting and estimation of the great-coach effect, using LSTM-based models for medal forecasting

Additional Awards Second-Class Scholarship, National College Student Mathematics Competition (Third Prize), Jilin Provincial Virtual Reality Competition (Second Prize), and SuperMap Cup University GIS Competition (Merit Award)

University Leadership

Class Monitor and Student Counselor Assistant, Jilin University